Output Form (OF)

Score: 4/5 — Minor gaps | Confidence: high

Benchmark: *DRBENCHMARK* | ***Context:*** *EU Pharmaceutical Regulatory NLP — French*

Output Form definition: Whether the expected output modality matches regional deployment needs and accessibility.

> Each section below is followed by an evidence table. Three types of evidence are cited: **[QN]** — verbatim quotes extracted from the benchmark paper; **[WEB-N]** — web sources consulted during assessment; **[DN]** / **[DATASET-DN]** — sampled datapoints from the benchmark dataset, with an interpretation note.

Justification

Output form is well-aligned. The deployment consumes label and score outputs in modalities the benchmark produces: IOB2 sequence labels via SeqEval [Q52], multiple-choice selections, multi-label classification labels, and continuous similarity scores [Q54]. Models predict only the first token label per word for tokenizer-agnostic evaluation [Q53]. Hyperparameters are documented [Q49, Q99] and best models saved on validation metrics [Q100]; results averaged over four runs with Student's t-test [Q51]. No output representation mismatch exists. Two minor concerns: (1) the EDRM metric [Q54] is calibrated for DEFT-2020 task design and has no documented regulatory calibration — this is a calibration concern that cuts across to OO; (2) compute requirements (~2,500 GPU hours [Q93], 36.9 kgCO2eq [Q94]) raise reproducibility concerns for resource-constrained users [Q92], though HuggingFace toolkit and documented hyperparameters [Q49] mitigate this. The output-form modality match is strong, justifying score 4 rather than 5.

ID	Evidence
Q52	“To ensure a fair and consistent comparison among systems for sequence-to-sequence tasks such as POS tagging and NER, we chose the SeqEval (Nakayama, 2018) metric in conjunction with the IOB2 format and the training of all the models to predict only the label on the first token of each word as mentioned by Touchent et al. (2023).” (p.6)
Q54	“For STS tasks, the models' performance was assessed using two metrics: (1) the Spearman correlation, and (2) the mean relative solution distance accuracy (EDRM), as defined by the original authors of the DEFT-2020 dataset (Cardon et al., 2020).” (p.6)
Q53	“It provides a tokenizer-agnostic evaluation and mitigates any correlation between models' performances and the tokenization process.” (p.6)
Q49	“For further guidance, we have integrated all the training details, including hyperparameters, in Appendix A. This information will help users to reproduce and customize the experiments conducted with DrBenchmark.” (p.5)
Q99	“For the experiments, we utilize the following hyperparameters that yield optimal performance from the models.” (p.13)
Q100	“To mitigate overfitting, we locally save the best model based on its validation metric.” (p.13)
Q51	“All the models are fine-tuned regarding a strict protocol using the same hyperparameters for each downstream task. The reported results are obtained by averaging the scores from four separate runs, thus ensuring robustness and reliability. We also report statistical significance computed using Student's t-test.” (p.5)
Q93	“We utilized approximately 2,500 hours on V100 GPUs from the Jean-Zay supercomputer to complete the quantitative study.” (p.9)
Q94	“According to the Jean Zay supercomputer documentation, the total environmental cost is estimated to be equivalent to 647,500 Wh or 36.9 kgCO2eq/kWh, based on the carbon intensity of the energy grid mentioned in the BLOOM environmental cost study conducted on Jean Zay (Luccioni et al., 2022).” (p.9)
Q92	“Although the benchmark is easily reproducible and customizable, it required a substantial amount of computational power to execute all runs.” (p.9)

Strengths

- IOB2 sequence labels via SeqEval are a deployment-compatible NER output format [Q52]
- Tokenizer-agnostic evaluation by predicting first-token label per word [Q53]
- Continuous STS scores with Spearman correlation provide compatible output modality even if scoring rule mismatches [Q54]
- Reproducibility supported by documented hyperparameters [Q49, Q99] and best-model saving on validation metric [Q100]

- Statistical significance reported via Student's t-test over four runs [Q51]

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Checklist

Checklist item definitions:

ID	Assessment Question
OF-1	Does output modality match regional deployment needs?
OF-2	Assess text-to-speech availability for speech output
OF-3	Consider literacy rates and accessibility requirements
OF-4	Document form mismatches harming external validity

Checklist responses:

OF-1: Output modality matches: deployment consumes label outputs (NER spans, classification decisions) and continuous scores (STS for safety-warning equivalence). Benchmark produces both [Q52, Q54]. No modality mismatch.

OF-2: Not applicable — deployment is text-only document management with no speech-output requirement [infrastructure_notes interface_modality].

OF-3: Target population is regulatory affairs specialists, pharmacologists, and legal experts — a professional cohort with full literacy. No accessibility requirement diverges from general French professional norms [target_population description].

OF-4: Form mismatches harming external validity: minimal at the representation level. The substantive concern (EDRM and Spearman correlation are not calibrated for regulatory equivalence detection) is properly attributed to OO (scoring rule), not OF (representation).

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